

# A Noise-Explicit POMDP Framework for Structural Health Monitoring

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**Abstract** Structural Health Monitoring (SHM) supports cost-effective maintenance by enabling condition-based decision-making, but its effectiveness is often limited by non-stationary and time-varying sensor noise. Conventional Partially Observable Markov Decision Process (POMDP) formulations typically assume stationary observation models, which can lead to overconfident state estimation and locally optimal maintenance policies. This study proposes a noise-aware POMDP framework that incorporates time-dependent observation uncertainty through state augmentation. By allowing the observation model to vary over time while preserving the Markov property, the approach captures temporal variations in sensing noise without requiring prior knowledge of its evolution. The framework is demonstrated on a vibration-based SHM case study of a benchmark beam structure. Results show that the proposed method improves robustness to time-varying noise, leading to more reliable state estimation and improved long-term decision performance compared with conventional stationary-observation POMDP models.

**Keywords:** Structural health monitoring; POMDP; time-varying observation noise; decision-making under uncertainty; infrastructure maintenance; vibration-based damage detection.

## Introduction

Inspection and maintenance planning determine actions required to retain a system in, or restore it to, an acceptable condition. Optimal maintenance schedules aim to ensure the desired performance at the lowest possible cost [1]. Structural health monitoring (SHM) supports this process by continuously monitoring structural behavior using sensor-based systems, extracting damage-related information from the collected data, and estimating the structural condition [2]. Given that structural conditions are typically not directly observed and inferred from noisy monitoring data, inspection and maintenance planning can be formulated as Partially Observable Markov Decision Process (POMDP) based on stochastic models and uncertain structural data. In this framework, the structural condition is modeled as a hidden state, while

inspections or monitoring provide noisy observations that update the belief state and guide maintenance decisions. Prior studies [3, 4] have successfully applied POMDPs to integrate inspection, monitoring, and maintenance actions under uncertainty.

Most existing POMDP-based approaches assume stationary transition and observation models, which is often unrealistic in SHM. In practice, deterioration rates may evolve over time [5], and sensor noise can vary due to environmental and operational conditions [6]. While prior work has addressed time-varying transitions through state augmentation [4], time-varying observation models remain largely unexplored. This is critical for vibration-based monitoring, where damage features may vary due to environmental effects rather than damage [6], potentially leading to incorrect condition inference and suboptimal decisions.

Some studies have considered incomplete or uncertain observations in POMDPs, such as missing data [7] or latent-state inference using Bayesian non-parametric methods [8]. However, these approaches typically assume stationary observation models or treat uncertainty as stochastic, without explicitly modeling time-varying observation noise. To our knowledge, no existing framework explicitly learns non-stationary observation uncertainty within POMDPs.

To address this limitation, this study introduces a time-augmented POMDP formulation that captures time-dependent observation uncertainty. The key idea is to augment the structural state with a time index. This enables the model to capture systematic temporal variations in sensing noise within the POMDP framework. Maintenance decisions are then obtained through online planning based on the updated belief state. The proposed approach is evaluated through a numerical case study on a two-span bridge beam structure.

This work makes three main contributions. First, we introduce a time-augmented POMDP formulation that incorporates time-varying observation uncertainty in SHM decision problems. Second, we validate the approach on a vibration-based beam case study. Third, we show that modeling time-varying observation uncertainty improves maintenance robustness under changing noise.

## 1. POMDPs

Infrastructure maintenance under uncertainty can be formulated as a POMDP, defined by the tuple  $\{S, A, \Omega, T, Z, R, \gamma\}$ , where  $S$  denotes deterioration states,  $A$  maintenance actions, and  $\Omega$  inspection observations. The transition model  $T(s' | s, a)$  captures stochastic deterioration and repair, while the observation model  $Z(o | s', a)$  represents imperfect sensing and inspection. Since the true state is not directly observable, decisions are made based on a belief state  $b$ , updated as  $b'(s') = \eta Z(o | s', a) \sum_{s \in S} T(s' | s, a) b(s)$ , where  $\eta$  is a normalization constant. The objective is to maximize the expected discounted reward over beliefs:

$$V(b) = \max_{a \in A} \left[ R(b, a) + \gamma \sum_{o \in \Omega} P(o | b, a) V(b') \right] \quad (1)$$

Although the belief space is continuous, the value function is piecewise-linear and convex structure and can be represented by a finite set of  $\alpha$ -vectors:

$$V(b) = \max_{\alpha \in \Gamma} b \cdot \alpha \quad (2)$$

where  $\Gamma$  denotes the set of  $\alpha$ -vectors representing the expected return of a specific conditional plan across all states, allowing the value of any belief to be computed as the maximum expected value over a finite set of such plans. This representation enables point-based solution methods that approximate the value function using a finite set of representative belief points. One of the most efficient algorithms is SARSOP [9], which focuses computation on the subset

of beliefs that are optimally reachable.

## 2. State Augmentation for Time-Varying Observation Models

In many infrastructure monitoring problems, the sensing process is nonstationary. The time variable can be viewed as a proxy for factors that influence sensing reliability but are not explicitly modeled, such as environmental conditions, sensor aging, or operational variability. To represent this nonstationarity while preserving the Markov structure of the decision process, we augment the latent state with time.

Let the latent damage state be  $d \in D$  and define the augmented state  $s = (d, t) \in S = D \times \mathcal{T}$ , where  $t$  represents the time index. The observation model becomes time-dependent:

$$Z(o \mid s, a) = P(o \mid d, t, a).$$

By conditioning observations on time, the model can represent systematic changes in sensing uncertainty without explicitly modeling each underlying source.

**Optimality preservation.** Augmenting the state with time increases the state dimension but does not restrict the set of feasible policies. Any policy that ignores time can still be represented in the augmented formulation by simply not conditioning decisions on  $t$ . Therefore, the optimal value in the augmented model cannot be worse than that of the original stationary formulation [10]. In this sense, state augmentation preserves optimality while enabling additional modeling flexibility.

**Robustness to imperfect time-dependent models.** The effectiveness of the augmented representation does not require the time-dependent observation model to be exact. It is sufficient that the model captures part of the temporal variation in sensing noise.

To illustrate this idea, consider a binary observation model where  $o_t = 1$  indicates a damage alarm. Suppose the true false-alarm probability for a healthy structure varies over time,  $P(o_t = 1 \mid d_t = 0, \tau_t) = \epsilon_{\tau_t}$ . If time is ignored, the observation model must use a stationary approximation based on the average noise level,  $\bar{\epsilon}$ . Now suppose a time-dependent estimate  $\hat{\epsilon}_{\tau_t}$  is used instead. Even if this estimate is imperfect, as long as it is closer to the true noise level than the stationary approximation,  $|\epsilon_{\tau_t} - \hat{\epsilon}_{\tau_t}| < |\epsilon_{\tau_t} - \bar{\epsilon}|$ , the resulting posterior belief will better explain the current observation.

In practice, this condition need not hold at every time step. It is sufficient that the time-dependent model captures systematic temporal variation in sensing uncertainty so that, on average, it provides a closer approximation to the true sensing process than a stationary model. In many SHM applications, sensing noise is influenced by environmental and operational conditions, making time-dependent modeling a practical way to reduce observation errors.

## 3. Experiment Setup

### 3.1 Structural Model and Decision Process

A numerical experiment is conducted on a two-span bridge beam structure (TU1402) [11] as illustrated in Fig.1. The decision process is over a 100-year planning horizon with a discount factor  $\gamma = 0.97$ . The system state is defined as  $s = (t, d)$ , where  $t \in \{0, \dots, 99\}$  is the time index and  $d \in \{0, 1, 2, 3\}$  represents the damage level (healthy (H0), minor damage (D1), major damage (D2), and failure (D3)). At each decision epoch, an action  $a = (i, r)$  is selected, consisting of an inspection decision  $i \in \{\text{no inspection (N)}, \text{inspection (I)}\}$  and

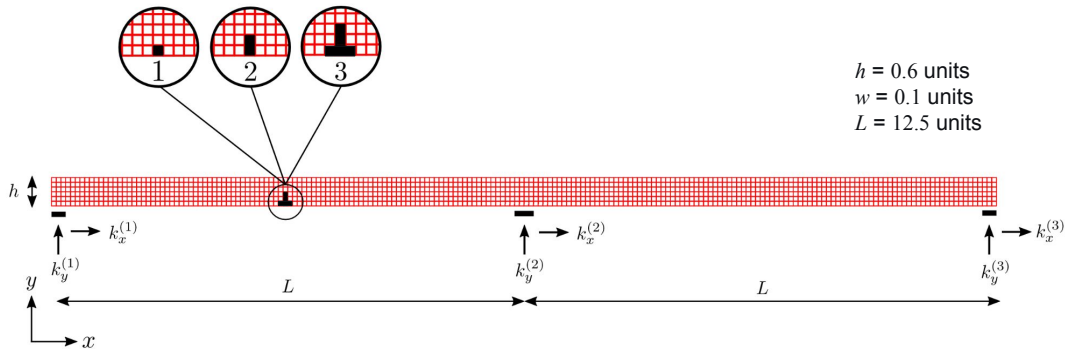


Fig. 1. Finite element model of the benchmark structure.

a repair decision  $r \in \{\text{no repair (N)}, \text{minor repair (R1)}, \text{major repair (R2)}\}$ . Damage evolution is modeled as a Markov transition process conditioned on the repair action. The natural deterioration process (no repair) is described by

$$P(d'|d, r = \text{no repair}) = \begin{bmatrix} 0.689 & 0.300 & 0.010 & 0.001 \\ 0 & 0.595 & 0.400 & 0.005 \\ 0 & 0 & 0.600 & 0.400 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

where rows correspond to the current damage state  $d$  and columns to the next state  $d'$ . Minor repair partially restores the structure, returning it to the healthy state with probability 0.9 and to minor damage from major damage with probability 0.8, while having no effect on the failed state. Major repair fully restores the structure to a healthy condition from any damaged state.

The true damage state is not directly observable; instead, observations  $o_t \in \{o^0, o^1, o^2, o^3\}$  are obtained from vibration-based features derived from the numerical model. Measurement uncertainty is simulated by adding Gaussian noise to the modal frequencies and mode shapes of the finite element model. For each noisy realization, damage-sensitive features are computed, including the log-relative frequency change and the modal assurance criterion (MAC) with respect to the health state. The feature vectors are standardized and projected onto the first principal component (PC1) via principal component analysis, yielding a scalar damage indicator. The PC1 scores are then clustered into four observation symbols using quantile-based thresholds. The discretizer is calibrated using noisy samples generated with frequency noise level  $\sigma_f = 0.08$  and mode-shape noise level  $\sigma_\phi = 0.06$ .

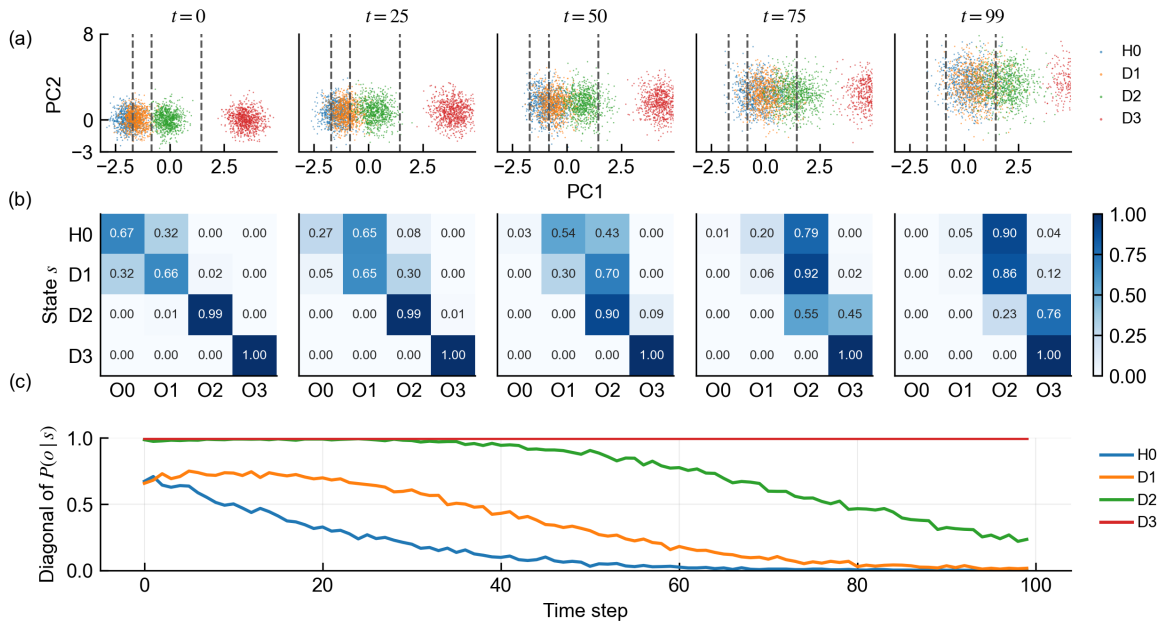
To quantify the effect of state augmentation under different noise levels and patterns, two sensor observation settings are considered. The first is a time-invariant model, where the noise level remains constant and matches that used for the discretizer calibration. The second is a time-variant model representing sensor degradation, where the noise increases linearly over the 100-year horizon from  $\sigma_f = 0.08$  to  $\sigma_f = 0.24$  for frequencies and from  $\sigma_\phi = 0.06$  to  $\sigma_\phi = 0.10$  for mode shapes. Fig. 2 illustrates the effect of time-variant sensor noise on the observation model. As shown in Fig. 2(a), increasing noise leads to progressively more dispersed and overlapping clusters in the PC1–PC2 space, reducing separability between damage states. This degradation is reflected in Fig. 2(b), where the observation probability matrices  $P(o | s)$  lose their diagonal dominance over time, indicating a higher likelihood of misclassification, particularly among  $H0$ ,  $D1$ , and  $D2$ . The trend is further quantified in Fig. 2(c), where the true-positive rates (diagonal elements of  $P(o | s)$ ) decrease significantly for  $H0$  and  $D1$ , more gradually for  $D2$ , while remaining near unity for  $D3$ . Overall, the results highlight the progressive loss of observation reliability due to sensor degradation and its impact on state distinguishability.

The inspection observation model  $Z(o|s, \text{inspection})$  provides a more accurate assess-

ment of the true damage state and is defined by

$$P(o'|d, i = \text{inspection}) = \begin{bmatrix} 0.95 & 0.05 & 0 & 0 \\ 0.05 & 0.90 & 0.05 & 0 \\ 0 & 0.05 & 0.90 & 0.05 \\ 0 & 0 & 0.05 & 0.95 \end{bmatrix}.$$

The reward function captures maintenance costs and structural performance penalties. Inspection incurs a cost of  $c_i = -2.5$ , while minor and major repairs incur costs of  $c_m = -35$  and  $c_M = -50$ ; additional penalties are applied for degraded states ( $-10$  for  $D2$  and  $-1000$  for  $D3$ ) and for observation misclassification, including a false positive cost of  $-5$  and a severe false negative cost of  $-100$ .



**Fig. 2.** Time-variant observation model. (a) Observations at selected time steps ( $t = 0, 25, 50, 75, 99$ ) under increasing noise levels using the same discretizer. (b) Observation probability matrices  $P(o | s)$  for the sensor without inspection at the selected time steps. (c) Evolution of the true-positive rates (diagonal elements of  $P(o | s)$ ) for each damage state over time.

### 3.2 Policy Evaluation

To evaluate the effect of observation noise, both time-invariant and time-invariant observation scenarios are considered. In the time-invariant scenarios, the noise level remains constant over the planning horizon with different magnitudes, whereas in the time variant scenarios the noise increases linearly over the 100-year horizon to represent sensor degradation. The discretizer is calibrated using baseline noise levels  $\sigma_f = 0.08$  for modal frequencies and  $\sigma_\phi = 0.06$  for mode shapes. In all scenarios (Table 1), the observation features are constructed from the first 10 modal frequencies and 10 mode shapes.

Monte Carlo simulations estimate expected policy performance by sampling transitions and observations. Over a 100-year horizon, maintenance effort is measured by inspection count, structural performance by visits to the health ( $H0$ ) and failure ( $D3$ ) states, and state estimation uncertainty by mean belief entropy

**Table 1.** Observation noise scenarios considered in the experiments.

| Type           | Scenario | $\sigma_f$              | $\sigma_\phi$           |
|----------------|----------|-------------------------|-------------------------|
| Time-invariant | Static 1 | 0.01                    | 0.04                    |
|                | Static 2 | 0.08                    | 0.06                    |
|                | Static 3 | 0.16                    | 0.08                    |
|                | Static 4 | 0.24                    | 0.10                    |
|                | Static 5 | 0.30                    | 0.12                    |
| Time-variant   | Time 1   | 0.08 $\rightarrow$ 0.16 | 0.06 $\rightarrow$ 0.08 |
|                | Time 2   | 0.08 $\rightarrow$ 0.24 | 0.06 $\rightarrow$ 0.10 |
|                | Time 3   | 0.08 $\rightarrow$ 0.30 | 0.06 $\rightarrow$ 0.12 |

$$\bar{H} = \frac{1}{T} \sum_{t=1}^T H(b_t), \quad H(b_t) = - \sum_{s \in S} b_t(s) \log b_t(s),$$

where  $b_t(s)$  denotes the belief probability of state  $s$  at time  $t$ . Lower entropy indicates greater certainty in the state estimate. For each scenario, all metrics are reported as the sample mean and standard deviation across rollout simulations.

## 4. Results and Discussion

Optimal policies were computed using the SARSOP solver [9], with statistics for both observation models summarized in Table 2. The time-variant formulation demands greater computational effort, requiring more backups and generating more  $\alpha$ -vectors, because its augmented state space increases the set of reachable beliefs. However, this yields a more precise policy approximation, evidenced by a smaller gap between the lower (LBound) and upper (UBound) bounds.

**Table 2.** Summary of POMDP solver performance under time-invariant and time-variant observation settings.

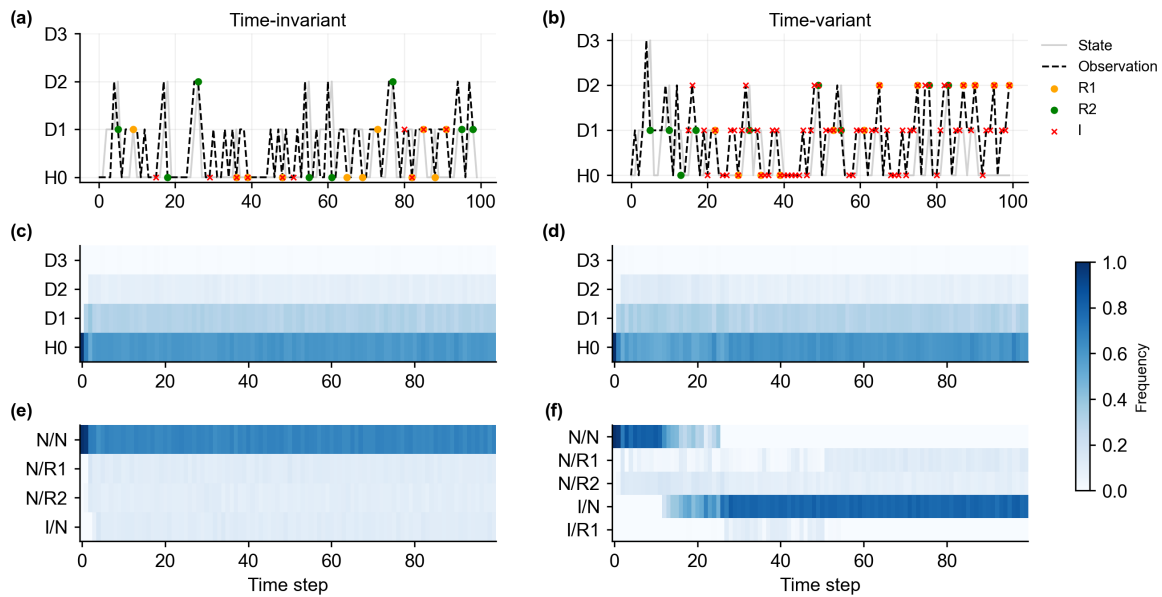
| Setting        | Time (s) | Trial | Backup | LBound   | UBound  | Precision | Alphas | Beliefs |
|----------------|----------|-------|--------|----------|---------|-----------|--------|---------|
| Time-invariant | 504.13   | 5648  | 252277 | -288.249 | -276.29 | 11.96     | 68     | 30407   |
| Time-variant   | 1136.25  | 9405  | 443083 | -295.45  | -291.48 | 3.97      | 604    | 28008   |

### 4.1 Policy Behavior

Figure 3 illustrates representative rollouts and action frequencies obtained from 1000 Monte Carlo simulations over a 100-year horizon. When observation noise is low, both policies exhibit similar behavior, indicating that time augmentation does not significantly affect decisions when sensing reliability is high.

As observation noise increases, the policies diverge. The time-variant policy conducts inspections more frequently to reduce uncertainty in the state estimate. In contrast, the time-invariant policy performs fewer inspections because the stationary observation model underestimates the degradation of sensing reliability over time. Consequently, the time-invariant policy tends to rely more heavily on repair actions without additional information. Another notable difference is the coordination of inspection and repair decisions. The time-invariant policy typically separates these actions, performing inspections before repairs or repairing directly. In contrast, the time-variant policy occasionally combines inspection with minor repair within the same decision step. This behavior reflects the policy's attempt to reduce uncertainty while simultaneously mitigating potential structural deterioration. However, even

when observations suggest health or minor damage, the policy may still repair, because partial observability can leave the belief state with substantial probability on higher-damage states.



**Fig. 3. Policy behavior under static and time-varying observation noise.** (a–b) Example rollouts showing true damage state (solid) and observations (dashed) over 100 years; markers indicate minor repair (R1), major repair (R2), and inspection (I). (c–d) State visitation frequencies from 1000 Monte Carlo rollouts. (e–f) Action selection frequencies across the same rollouts. N = No Action

#### 4.2 Performance Across Noise Scenarios

Table 3 summarizes the performance of the two policies across different observation-noise scenarios. Under low noise levels (Static 1–2), the time-invariant policy achieves slightly higher reward because the stationary observation assumption is approximately valid and additional inspections provide limited benefit.

**Table 3.** Performance comparison of time-invariant and time-variant policies across observation-noise scenarios. Values are means with standard deviations in parentheses from Monte Carlo simulations.

| Scenario | Policy         | Reward          | No. Inspection | Time in H0  | Time in D3 | Belief entropy |
|----------|----------------|-----------------|----------------|-------------|------------|----------------|
| Static 1 | Time-invariant | -364.57(162.44) | 23.19(1.68)    | 60.03(5.90) | 0.17(0.42) | 0.47(0.02)     |
|          | Time-variant   | -390.86(195.95) | 64.26(2.94)    | 58.66(5.35) | 0.19(0.44) | 0.33(0.02)     |
| Static 2 | Time-invariant | -403.84(184.08) | 11.10(2.84)    | 60.26(5.81) | 0.20(0.44) | 0.48(0.02)     |
|          | Time-variant   | -413.28(181.89) | 65.24(2.76)    | 61.40(5.19) | 0.18(0.42) | 0.33(0.02)     |
| Static 3 | Time-invariant | -625.72(167.48) | 1.02(1.09)     | 69.62(6.89) | 0.22(0.44) | 0.62(0.02)     |
|          | Time-variant   | -508.35(172.97) | 65.28(2.69)    | 63.17(5.32) | 0.18(0.42) | 0.36(0.02)     |
| Static 4 | Time-invariant | -631.63(217.74) | 0.71(0.83)     | 70.08(5.91) | 0.25(0.52) | 0.59(0.01)     |
|          | Time-variant   | -502.51(233.34) | 66.02(2.62)    | 62.89(5.39) | 0.20(0.46) | 0.35(0.02)     |
| Static 5 | Time-invariant | -729.46(311.62) | 0.00(0.00)     | 69.13(5.95) | 0.44(0.84) | 0.63(0.01)     |
|          | Time-variant   | -545.69(236.80) | 66.04(2.62)    | 62.90(5.07) | 0.22(0.50) | 0.36(0.02)     |
| Time 1   | Time-invariant | -432.78(172.11) | 9.78(2.38)     | 61.78(6.65) | 0.17(0.40) | 0.56(0.02)     |
|          | Time-variant   | -420.00(201.46) | 64.77(2.89)    | 60.04(5.17) | 0.20(0.45) | 0.33(0.02)     |
| Time 2   | Time-invariant | -484.30(168.76) | 6.24(2.10)     | 64.56(6.90) | 0.26(0.53) | 0.58(0.02)     |
|          | Time-variant   | -414.31(197.35) | 64.83(2.97)    | 60.20(5.33) | 0.20(0.45) | 0.34(0.02)     |
| Time 3   | Time-invariant | -522.25(177.86) | 7.04(1.88)     | 65.29(6.37) | 0.31(0.59) | 0.59(0.01)     |
|          | Time-variant   | -414.93(181.92) | 65.20(2.75)    | 60.45(5.37) | 0.20(0.46) | 0.34(0.02)     |

As observation noise increases (Static 3–5), the advantage shifts to the time-variant

policy. The stationary model fails to capture the degradation of sensing reliability, leading the time-invariant policy to significantly reduce inspection actions. This results in higher belief entropy and less reliable state estimates. In contrast, the time-variant policy maintains frequent inspections, which improve state estimation and yield higher overall rewards. A similar pattern is observed in the time-varying noise scenarios (Time 1–3). The time-variant policy consistently achieves higher rewards and lower belief entropy, indicating more reliable condition inference. Although it spends slightly less time in the healthy state ( $H0$ ), it also reduces the time spent in the failure state ( $D3$ ), suggesting improved risk management under increasing observation uncertainty.

## 5. Conclusion

This study investigated the impact of time-varying observation uncertainty in POMDP-based frameworks for structural health monitoring (SHM). Current approaches often assume stationary transition and observation models; however, this assumption may not hold in practice, as sensing reliability and environmental conditions can change over time. To address this, we proposed a time-augmented POMDP formulation that captures temporal variation in the observation process by augmenting the latent state with time. The approach was demonstrated through a vibration-based monitoring case study on a benchmark beam structure. The results show that explicitly modeling time-dependent observation uncertainty leads to more robust maintenance policies when sensing reliability changes.

Limitations include the focus on a simplified beam structure and a specific representation of uncertainty that may not capture all real-world variability. Despite these limitations, the results demonstrate that incorporating time-dependent observation effects can substantially improve the reliability of POMDP-based SHM decision frameworks. Future research may explore more flexible representations of time-varying observation models, integration with adaptive learning techniques, and validation using long-term field monitoring data. Such developments would further enhance the practical applicability of decision-making frameworks for infrastructure monitoring.

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